

(4) CONVEX AND CONCAVE LENSES**Aim**

Determination of the focal length of,

- (a) a convex lens by,
 - I. the magnification method,
 - II. the conjugate points method and
- (b) a concave lens by non-parallax method.

Materials and Apparatus

Convex lens, concave lens, filament bulb, meter ruler, glass ruler, Vernier caliper, ground glass, wire mesh, white screen, 2 optical pins, a plane mirror strip, 3 stands and 2 lens holders.

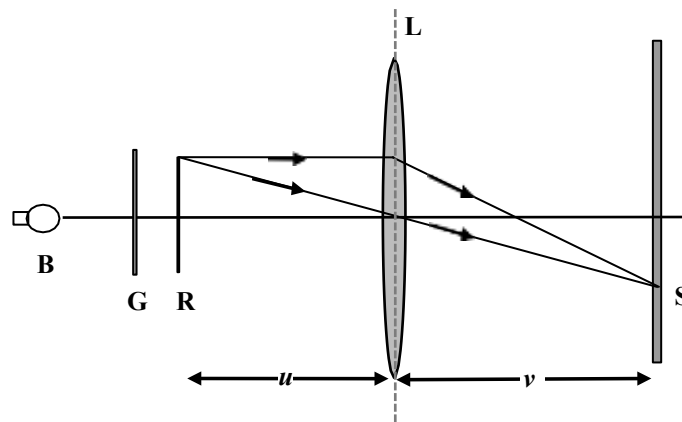
Theory**Part (a) (I)**

Fig. 1: Apparatus setup for determining the focal length of a convex lens (magnification method) where, B: Filament bulb, G: Ground glass, R: Glass ruler, L: Convex lens, S: White Screen, U : Object distance, V : Image distance.

From the lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \quad (1)$$

The linear magnification (M) of a convex lens can be written as,

$$M = \frac{v}{u} \quad (2)$$

Using the Cartesian sign convention for a real image of a convex lens, equations (1) and (2) can be reduced to,

$$\frac{1}{M} = \frac{u}{f} - 1 \quad (3)$$

where, f is the focal length.

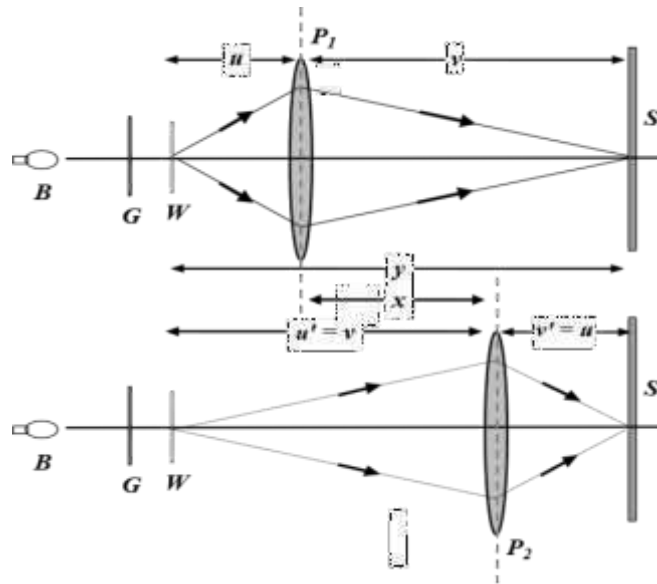
Part (a) (II)

Fig. 2: Conjugate positions (P_1, P_2) of a convex lens
where, W: Wire mesh, u' : Object distance, v' : Image distance

From Figure 2, $y = u + v$ and $x = v - u$

Then, $v = \frac{x + y}{2}$ and $u = \frac{y - x}{2}$

Substituting u & v in the lens formula gives, $f = \frac{y^2 - x^2}{4y}$ (4)

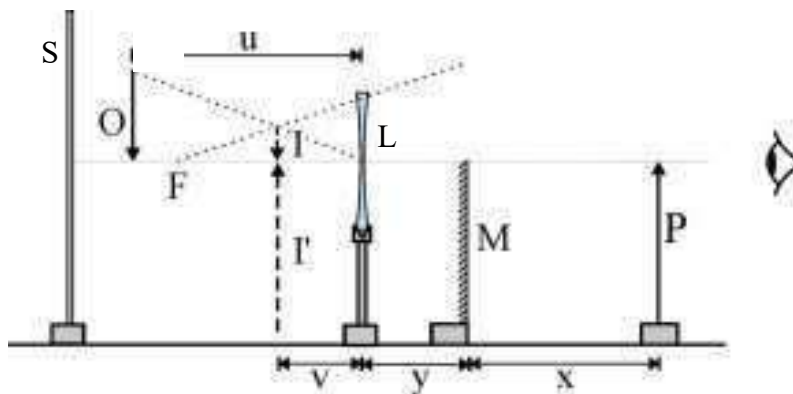
Part (b)

Fig.3: Apparatus setup for determining the focal length of a concave lens (non-parallax method) where, O,P: Optical pins, I,I': Images, S: Screen, M: Mirror, u : Object distance, v : Image distance, y : Distance between mirror and lens, x : Distance between pin (P) and mirror

For a concave lens, if u is the object distance, v is the image distance and f is the focal length,

According to the common formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

For concave lens,

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} \quad (5)$$

$$x = y + v \quad (6)$$

Procedure

Part (a) (I)

- Obtain **approximate** focal length (f) using an appropriate method with the given apparatus.

Q: Why should you find the approximate focal length?

- Set up the apparatus as shown in Figure 1.
- Place the glass ruler (object) in front of the lens and move the screen until a clear image of the glass ruler is obtained.

Q: Is there a specific range to place the lens?

Q: How do you measure the magnification (M)?

- Measure the object distance u .
- Repeat the above procedure in order to determine f , graphically.
- Estimate the uncertainty of f .

Part (a) (II)

- Obtain a real image of the wire mesh on the screen and mark the P_1 position of the lens.
- Find the other conjugate position P_2 .
- Obtain the required measurements to calculate the focal length of the convex lens.

Part (b)

- Arrange the setup as shown in figure 3
- Place the plane mirror strip (M) as shown in the figure on the side opposite to that of the object (O) and below the principal axis of the lens covering one half of the lens
- Fix the second optical pin P on a stand so that the tip of the pin touches the principal axis of the lens.
- Find the place where two images coincide with each other and get the necessary reading.

Q: How do you ensure that two images are aligned with each other?